

Reader: Over the intervals of finite length both bounded and unbounded functions may also be defined. Several illustrations of such functions are also shown in Fig. 15: the functions in cases (a) and (f) are bounded; the functions in cases (g), (h) and (i) are unbounded.

Author: You are right.

Reader: I note that in the cases that I have indicated the bounded functions are defined over the closed intervals $[-1, 1]$ for (a) and $[1, 2]$ for (f), while the unbounded functions are defined both over the open and half-open intervals $]1, 2[$ for (g), $]1, 2]$ for (h), and $[1, 2[$ for (i).

Author: This is very much to the point. However, you should bear in mind that it is possible to construct bounded functions defined over open (half-open) intervals, and unbounded functions defined over closed intervals.

Here are two simple illustrations:

Example 1: $y = x^2, 0 \leq x \leq 2$ **Example 2:** $y = \begin{cases} \frac{1}{x}, & 0 < x \leq 2 \\ 1, & x = 0 \end{cases}$

The graphs of these functions are shown in Fig. 16.

Reader: It seems that the boundedness (unboundedness) of a function and the finiteness of the interval over which it is defined are not interrelated. Am I right?

Author: Not completely. There is, for example, the following

Theorem: If a function is defined over a closed interval and if it is monotonic, the function is bounded. ↳ Imp

Reader: Obviously, the monotonicity of a function is determined similarly to the monotonicity of a numerical sequence.

Author: Yes, it is. Monotonic functions can be classified, as sequences, into nondecreasing and nonincreasing.

Definition: A function $y = f(x)$ is said to be nondecreasing over an interval D if for any x_1 and x_2 from this interval $f(x_1) \leq f(x_2)$ if $x_1 < x_2$. If, however, $f(x_1) \geq f(x_2)$, the function is said to be nonincreasing.